Digital logic & Design → (DLD)

Syllabus.

Mastery series.

corrupt Question.

1. GATE 2027 -2028.
2. Handwritten notes → Make your own notes
3. Extra → no of questions (PYQ's) ≃ 80%.
4. Patience →
5. minimum length 40-50.
6. PYQ's.
7. 2-3 doubt session. → Conversation.

Boolean Algebra ✓ Urdu lecture + notes + Same day Revise + PYQ's

Testseries → Topic wise test

Syllabus

Unit ①

① Basics (Signal , Dual, compliment, 1's comp, 2's comp, Logic system)

Boolean Algebra (Properties , theorem, SOP, POS, Standard form, minterm, maxterm, self dual Expression, → neutral function. → mutually Exclusive term).

Logic Gate · (NOT, OR, AND, NAND, NOR, Ex-OR, Ex-NOR)

NOT, OR, AND → Basic

NAND, NOR → universal logic gate

Ex-OR, Ex-NOR → Derived gate.

{ Svan suitable minimization.

K-map (2var, 3var, 4var, 5var)

(Prime implicant, Essential prime implicant)

(PI) (EPI)

(PI) ≃ 80%.

= Number System. (1's comp. 2's comp. unsigned nos., signed, BCD, Ex-3,)

Course + Revise → Mock.
(5/-10/-)

unit ② Combination ckt

→ Arithmetic circuits (HA, HS, FA, FS, parallel Adder, Carry look Ahead Adder),

→ MUX → Multiplexer and Demultiplexer (DEMUX)

→ Encoder & Decoder

→ Comparator.

→ parity Generator and checkers.

→ PROM's

unit ③ Sequential circuit

→ Latch (NAND latch, NOR latch)

→ Flip flops (S-R, J-K, D, T)

→ Shift Register (SISO, SIPO, PISO, PIPO)

→ Counters (Synchronous (Ring counter, Johnson Ring), Asynchronous Counter (up, down) (Binary, non Binary))

unit ④ — Data Converters.

- ADC
- DAC

unit ⑤ — logic families.

unit ⑥ — Semiconductor memories.

} Usual sort

p notes + PYQ's pdf.

④ State machines (moore , mealy m/c)

(overlapping detector
non-overlapping detector)

Basics

11 2

Basics

1's compliment:

A = 1000110 ⟶ 0111001 (1's comp)

0's ⟷ 1's

B = 11101 ⟶ 00010 (1's comp)

2's compliment:

(1) A ⟶ A' (1's comp) ⟶ A' + 1

2's compliment

A = 1 1 0 0 0 1 1

2's compliment —

A = 1100011

↓ 1's Compliment

0 0 1 1 1 0 0

+1

0 0 1 1 1 0 1 → 2's Compliment.

Shortcut:

A = 1 1 0 0 0 1 1

0 0 1 1 1 0 1

$$10000 \leftarrow$$

$$01000 \leftarrow$$

$A = 00001100\,1$ — 2's compliment ?

↓ 1's compliment

$$\begin{array}{r} 11110011 \\ +\ \ \ \ \ \ \ 1 \\ \hline 11110100 \end{array} \rightsquigarrow$$

Shortcut:

$A = 00001100\,1$ ←

$11110011 \leftarrow 1$

$\rightsquigarrow$ 11110100

Base = (2)

2 - 2 = 0

1 carry

A = 1 1 1 0 0 1 1 0 0 0 0

↓ 1's compliment

1 1 1 1

0 0 0 1 1 0 0 1 1 1 1

+ 1

0 0 0 1 1 0 1 0 0 0 0

Shortcut :

A = 1 1 1 0 0 1 (1) 0 0 0 0

0 0 0 1 1 0 1 0 0 0 0

(0, 1) → Base (2) → Binary Addition.

$$\begin{array}{r} {\scriptstyle 1} \\ +\ 0 \\ \hline 1 \end{array}$$

$$\begin{array}{r} 1\ \ {\scriptstyle 1} \\ +\ 1 \\ \hline 0 \end{array}$$

(2) - 2 = 0

(1) 9 + 3 → (12) - 10

= 2

$$\begin{array}{r} 9 \\ +\ 3 \\ \hline (12) \end{array}$$

Decimal Base (10)

$$\begin{array}{r} (1) \\ {\scriptstyle 1} \\ +\ 1 \\ \hline (1\ 1) \end{array} \rightarrow (3)$$

3 - 2 = 1

min bits

0 → 0

1 → 1

2 → 10

3 → 11

4 → 100

5 → 101

6 → 110

7 → 111

8 → 1000

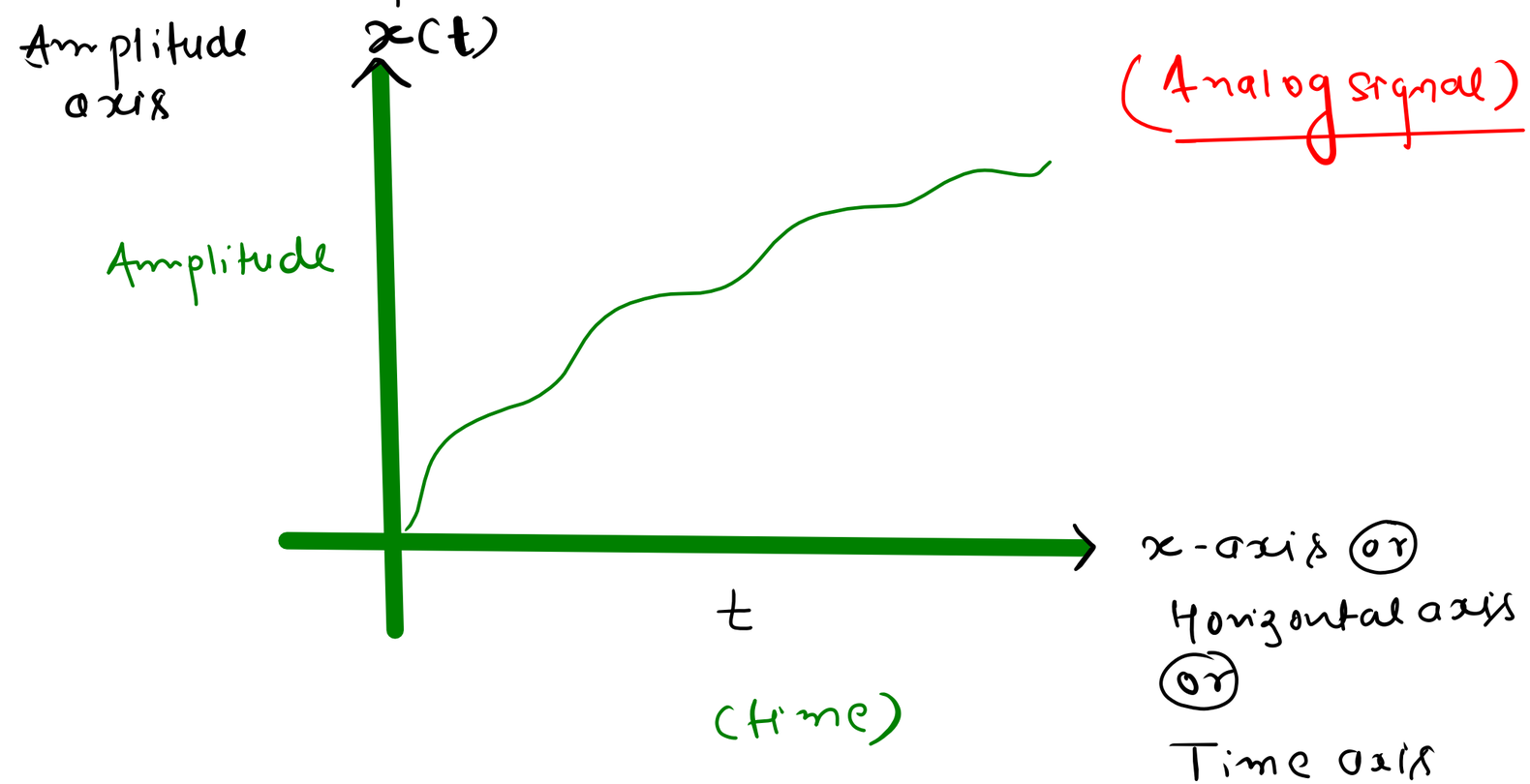
9 → 1001

10 → 1010

① Analog signal:

If the signal Amplitude can take infinite no of possibilities then it is called as Analog signal.

- Time → Continuous
- Amplitude → Continuous.



$= \{t=1, t=2, t=3\}$

3 values

finite.

$1 \leq t \leq 3$

Infinite values possible.

$1 \leq t \leq 2$

Infinite values possible.

time.

$t = 1.000001$

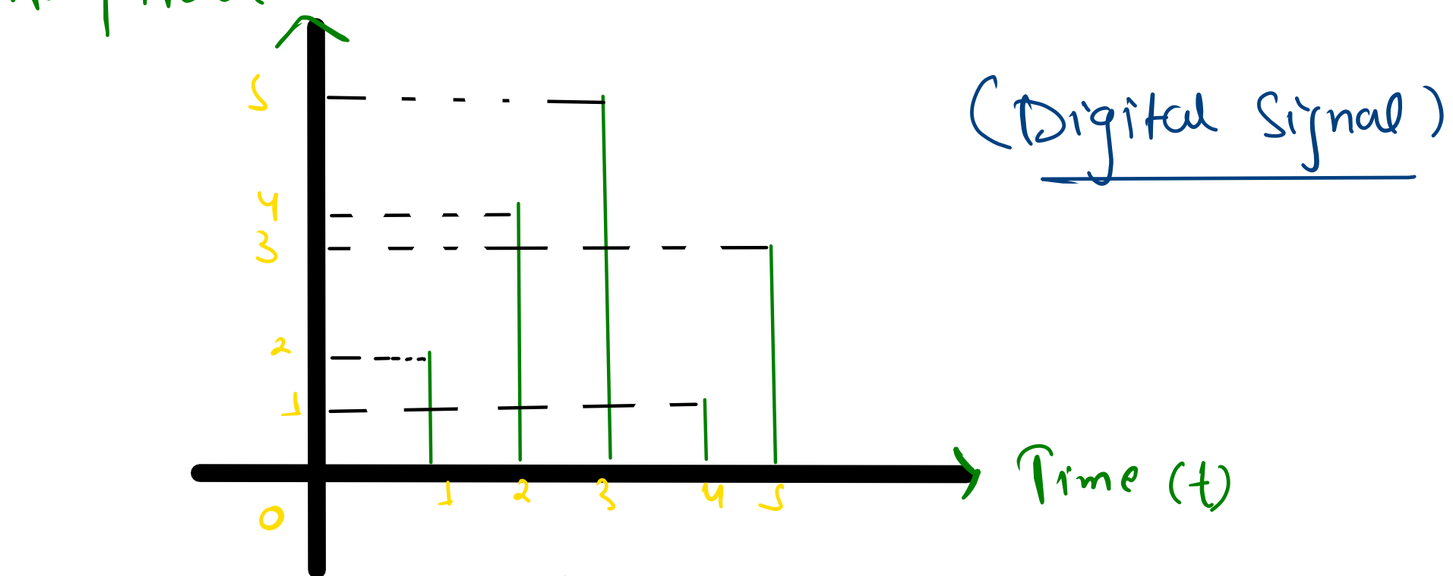
1.0000001

.
.
.

(2) **Digital Signal:** If the signal Amplitude can take only finite number of possibilities then it is called as Digital signal.

Time → Discrete,

Amplitude → Discrete.



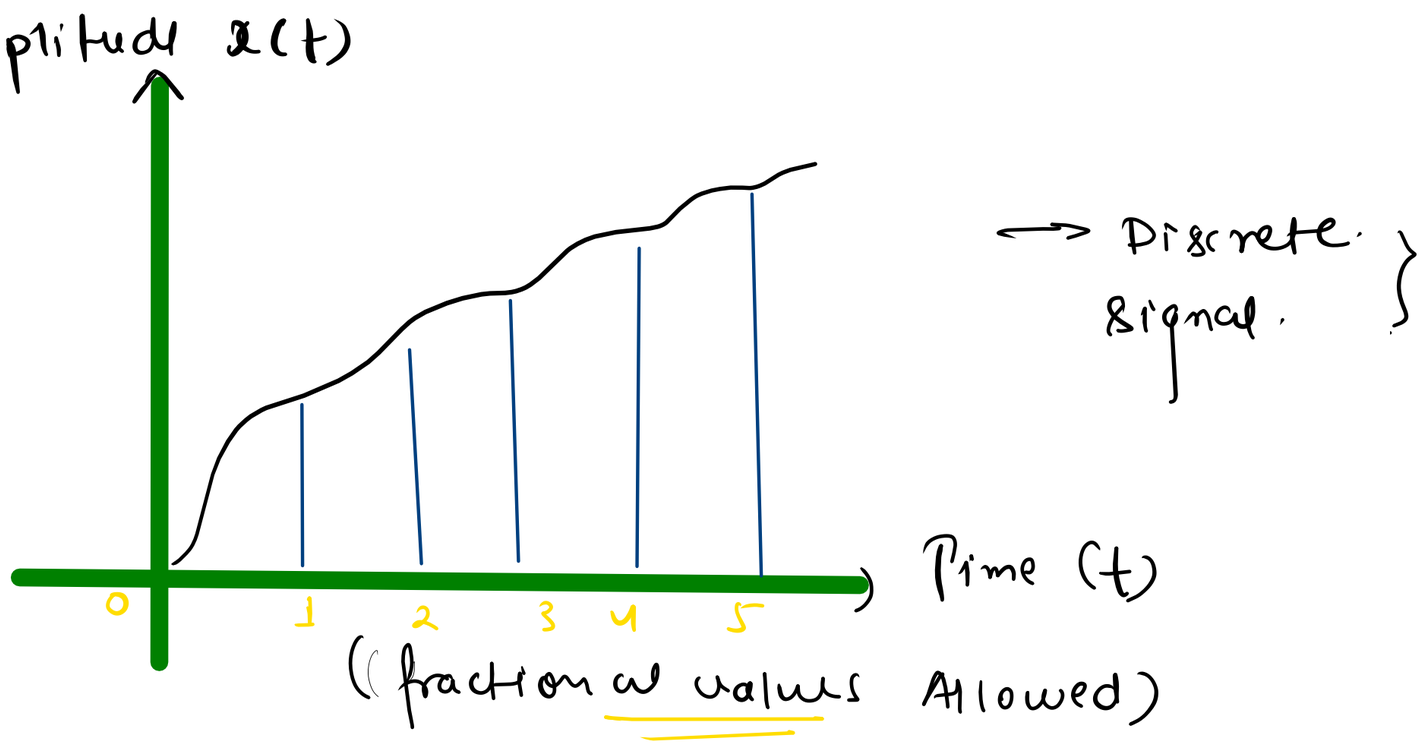
(Digital Signal)

(fractional values not allowed)

③ Discrete Signal :

Amplitude : Continuous.

Time : Discrete.

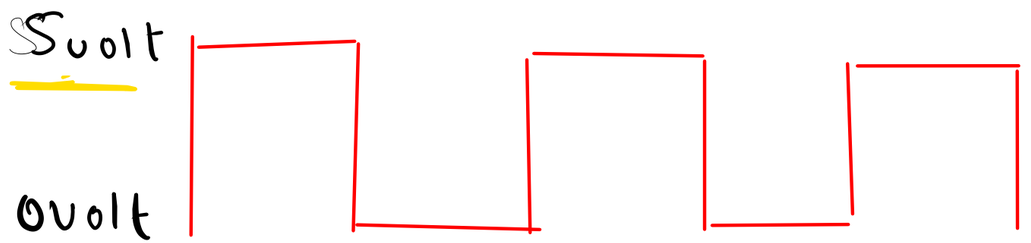


→ Discrete Signal.

((fractional values Allowed)

Amplitude	Time	Signal
Continuous	Continuous	Analog Signal
Continuous	Discrete	Discrete Signal
Discrete	Discrete	Digital Signal

Binary digital Signal. (Binary Signal)



function and Signal are same.

Note: ① If the signal takes only two possible Amplitudes then this is called as Binary digital signal.

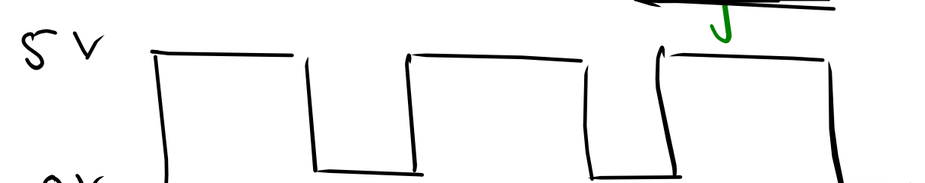
② The System which process the Analog signals is called as Analog System.

③ The System which process the digital signals is called as Digital System.

-10V , -3V

Logic Systems.

① positive logic system: High voltage corresponds to logic '1'.



{ +5V ----> logic '1'
0V ---> logic '0'

2V ------> logic '1'
-3V ------> logic '0'
-10V ----> logic '0'
-3V ----> logic '1'

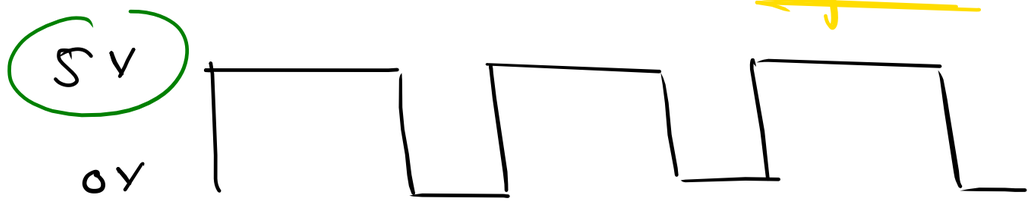
+ve logic system:

High voltage → logic '1'

low voltage → logic '0'

② Negative logic System!

High voltage corresponds to logic '0'



5V ------> logic '0'

0V ------> logic '1'

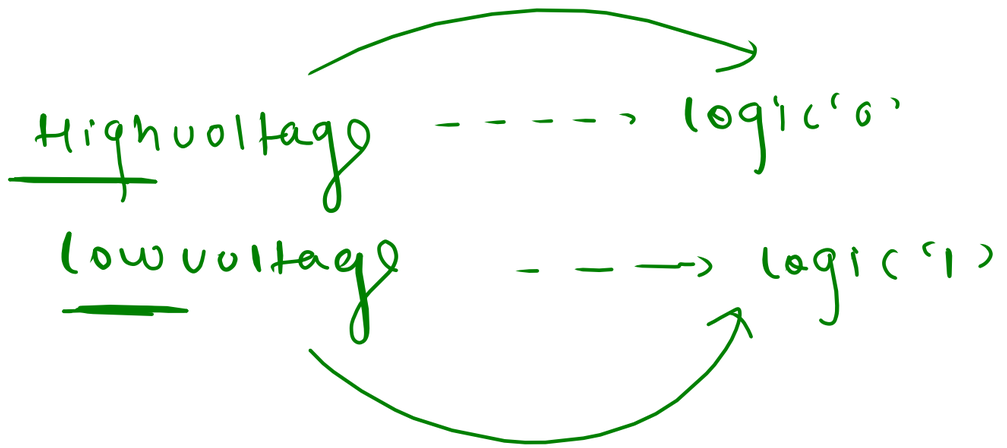
2V ------> logic '0'

-3V ------> logic '1'

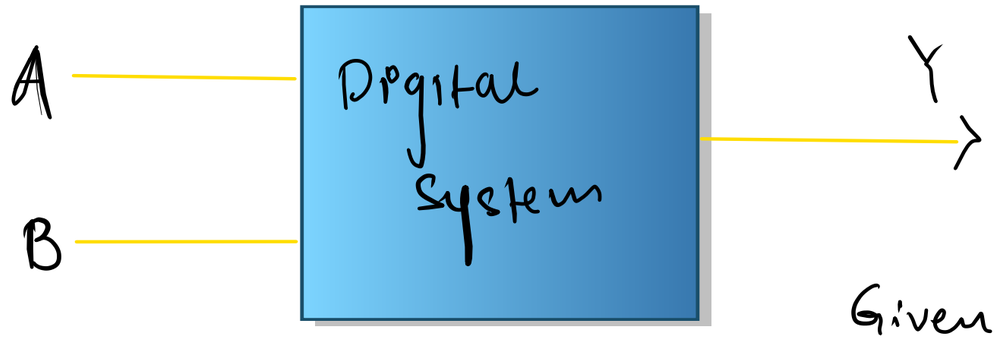
-3V ------> logic '0'

-10V ------> logic '1'

{ -ve logic system :



Example:



Given table:

A	B	Y
0V	0V	5V
0V	5V	0V
5V	0V	5V
5V	0V	0V

Highvoltage → logic '1'

+ve logic system.

A	B	Y
0	0	1
0	1	0
1	0	1
1	0	0

Highvoltage → logic '0'

-ve logic system.

A	B	Y
1	1	0
1	0	1
0	1	0
0	1	1

Duality

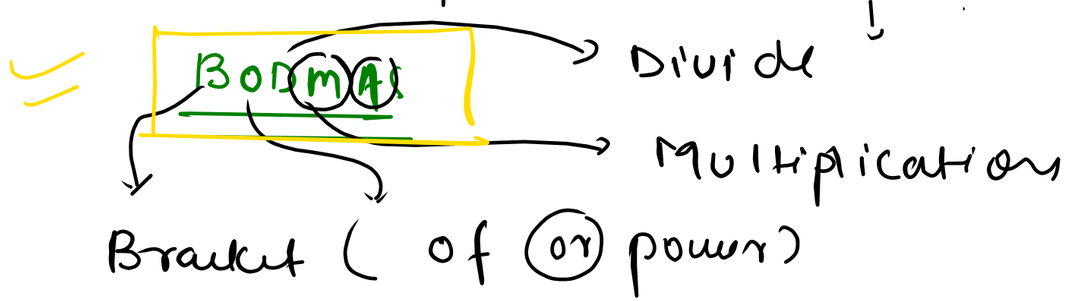
A positive logic system converted into negative logic system By using the concept of duality vice versa.

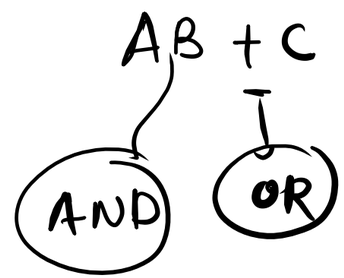
Steps to follow:

1. + (OR) ⟷ * (AND)
2. 0 ⟷ 1
3. Keep the variable as it is.

Ques① : Find the dual of the Expression $F = AB + C$

Soln:





$F = A \cdot B + C$

$F_d = (A + B) \cdot C$

Bracket ↑ AND ↓ OR

Ques (2): Find the Dual of the Expression $F = A(B+C)$

Sol^{n}:

$F = A.(B+C)$ ⟶ $A.(B(+)C)$

$F_d = A+(BC)$

B.C

Bracket > AND > OR

Boolean Algebra

- It's an Analysis tool that is used for analysing and designing of various digital system.
- The input vs o/p relationship in digital system is known as
  = Logic Expression.
  ↳ Boolean Expression.

A = 0 (or) 1

B = 0 (or) 1

OR operation · (+)

1). $A + 0 = A$

⇒ 2) $A + 1 = 1$

3) $A + A = A$

(4) $A + \bar{A} = 1$

OR

dual

AND operation (*)

1) $A \cdot 1 = A$

2) $A \cdot 0 = 0$

3) $A \cdot A = A$

(4) $A \cdot \bar{A} = 0$

AND

$1 + \text{Anything} = 1$ *

✓ (1) $A + 0 = A$, $A = 0$ or 1

$\bar{0} = 1$

$\bar{1} = 0$

Case (1): $A = 0$

$0 + 0 = 0 \rightarrow A$

Case (2): $A = 1$

$1 + 0 = 1 \rightarrow A$

(2) $A + 1 = 1$, $A = 0, 1$

Case (1): $A = 0$

$0 + 1 = 1$

Case (2): $A = 1$

$1 + 1 = 1$

(3) $A + A = A$

Case (1): $A = 0$

$0 + 0 = 0 \rightarrow A$

Case (2): $A = 1$

$1 + 1 = 1 \rightarrow A$

(4) $A + \bar{A} = 1$

Case (1): $A = 0$

$0 + \bar{0} = 0 + 1 \Rightarrow 1$

Case (2): $A = 1$

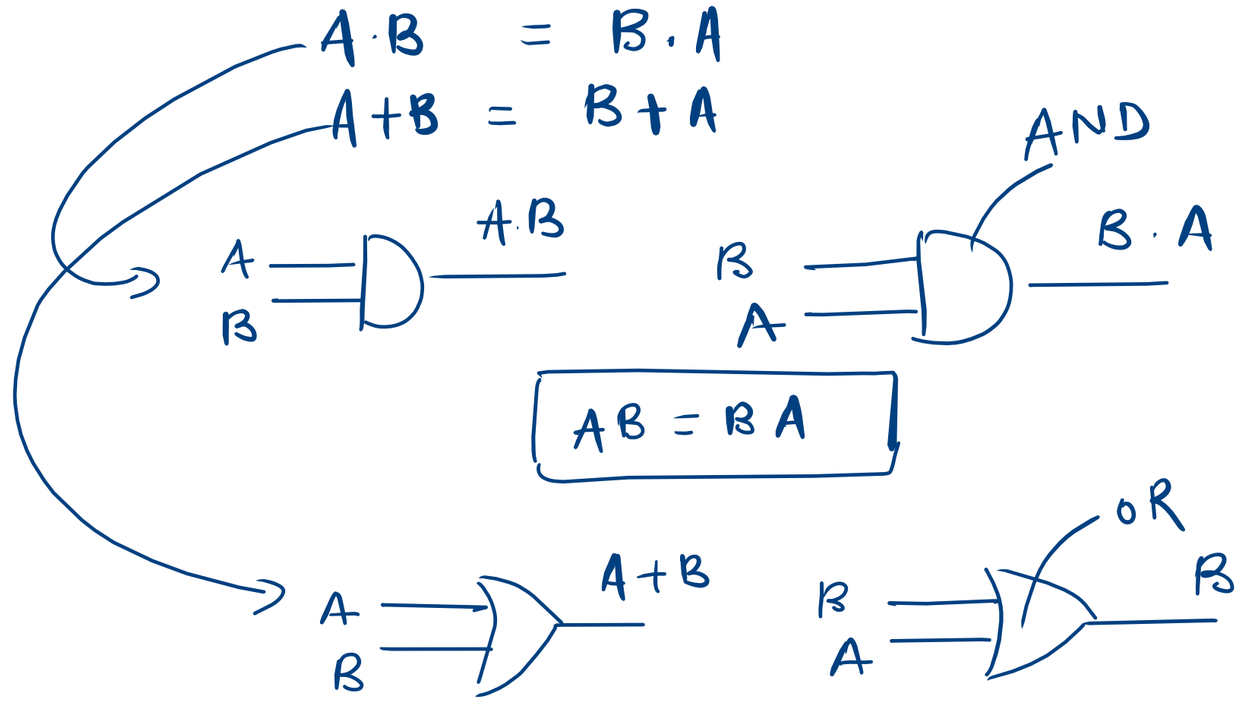
$1 + \bar{1} = 1 + 0 = 1$

Laws of Boolean Algebra

1). Commutative law:

$A \cdot B = B \cdot A$

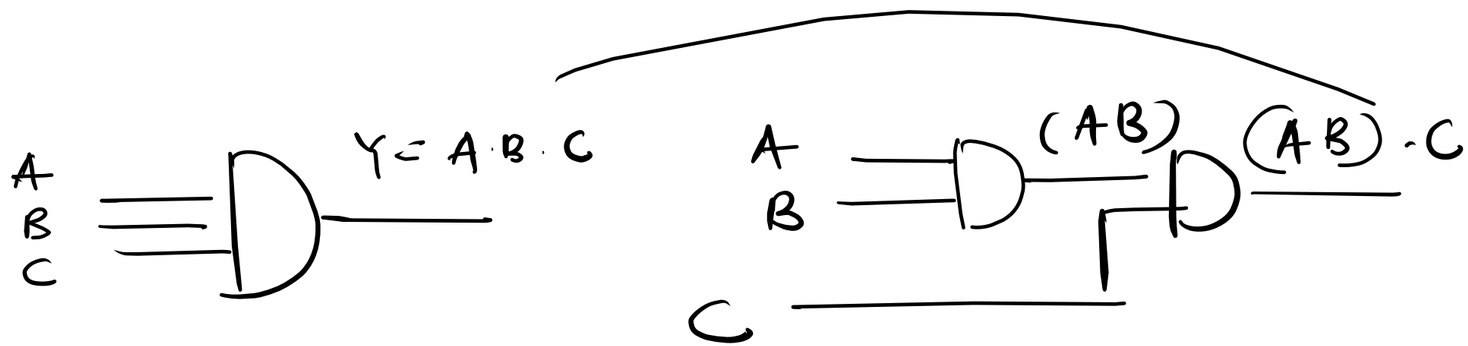
$A + B = B + A$

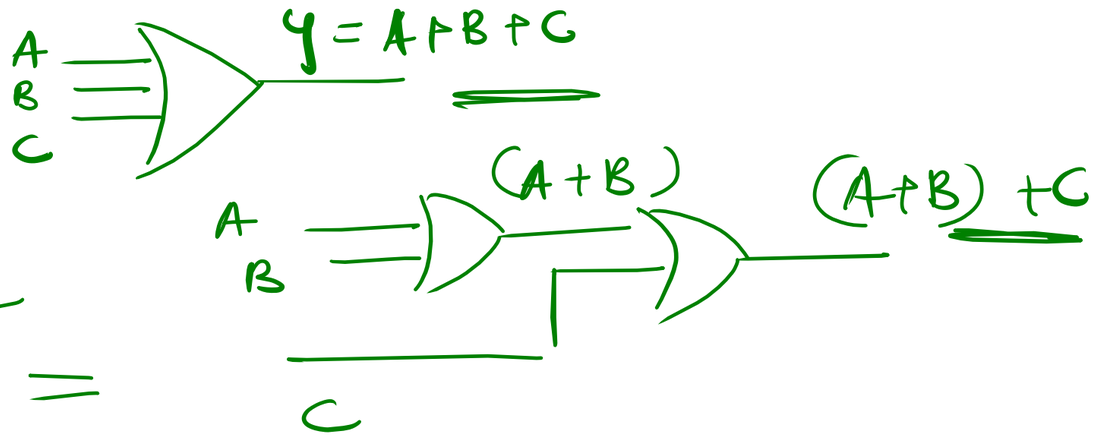


2). Associative law:

$A+B+C = (A+B)+C = A+(B+C) = (A+C)+B$

$A \cdot B \cdot C = (A \cdot B) \cdot C = A \cdot (BC) = (AC) \cdot B$





(3) **Distributive law:** (mingle property)

$$A \cdot (B+C) = AB + AC$$

$$A + (BC) = (A+B) \cdot (A+C)$$

RHS: $(A+B) \cdot (A+C)$

$$A \cdot A + AC + AB + BC$$

$$A + AC + AB + BC$$

$$A[1 + C + B] + BC$$

$$1$$

$$A \cdot 1 + BC \longrightarrow A + BC$$

④ D-morgan's law: (Compliment)

$$\overline{A \cdot B \cdot C \cdot D} = \bar{A} + \bar{B} + \bar{C} + \bar{D}$$

$$\overline{A + B + C + D} = \bar{A} \cdot \bar{B} \cdot \bar{C} \cdot \bar{D}$$

$\neq \bar{A}$ ✗ (A)

$$\overline{\overline{A \cdot B \cdot C}} \Rightarrow A \cdot B \cdot C$$

$$\overline{A \cdot B \cdot C} \rightarrow \overline{A \cdot B \cdot C} \rightarrow \bar{A} + \bar{B} + \bar{C}$$

(5) Transposition theorem: (T_1)

$$(A+B)\cdot(A+C) = A+BC$$

① ② ③ ④

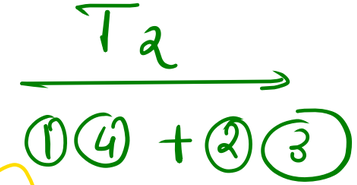
$A \cdot A + B \cdot C$

$A + BC$

(6) Transposition theorem (T-2):

$$(A+B)(\bar{A}+C) = AC + \bar{A}B$$

① ② ③ ④

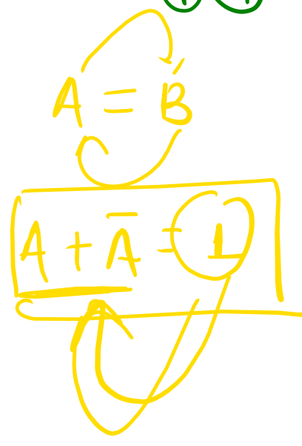


$(A+B)\,\bar{A}+C)$

$= A\cdot\bar{A} + AC + \bar{A}B + BC$

$= AC + \bar{A}B + BC$

$= AC + \bar{A}B + 1\cdot BC$



$1 \cdot BC = BC$

$1 + B = 1$

$$= AC + \bar{A}B + (A + \bar{A}) \cdot BC$$

$$= AC + \bar{A}B + ABC + \bar{A}BC$$

$$= AC(1 + B) + \bar{A}BC(1 + C)$$

$$= AC + \bar{A}B$$

Qus①: Minimize the following Expression $(x+y)(x+\bar{y})(\bar{x}+y)$

Soln:

$$F = (x+y)(x+\bar{y})(\bar{x}+y) \rightarrow (T_1)$$

$$F = (x \cdot x + y \cdot \bar{y})(\bar{x}+y)$$

$$= (x+0)(\bar{x}+y)$$

$$= x(x+y)$$

$$= x \cdot \bar{x} + xy$$

$$= xy$$

$$\begin{cases} (A+B)(A+C) = A+BC \\ (A+B)(\bar{A}+C) = AC+\bar{A}B \end{cases}$$

Qus ②: Minimize the Following.

① ② ③ ④

$(x+y+z)$ $(x+y+\bar{z})$

A A

$x+y+z\cdot\bar{z}$

$= x+y+0$

$= x+y$

Soln:

$A\cdot A+BC = A+BC$

T_1

①③ + ②④

$(x+y)\cdot(x+y)$

$A\cdot A = A$

$(x+y)$

Qus ③ :

$(\bar{A}+B)(\bar{A}+\bar{B})$

$\bar{A} + B\cdot\bar{B}$

$= \bar{A} + 0$

$= \bar{A}$

T_1

① ③ + ② ④

Qus ④ :

$A + \bar{A}B$

$(A+\bar{A})(A+\bar{B})$

$(A+\bar{B})$

Qus ⑤ :

$\bar{A} + A\bar{B}$

$(\bar{A}+A)(\bar{A}+\bar{B})$

$(\bar{A}+\bar{B})$

Qus: Find the Compliment of the Expression.

$$F = AB + C$$

Soln:

$$F_c = \overline{AB + C}$$

$$= (\overline{AB}) \cdot \overline{C}$$

$$= (\overline{A} + \overline{B}) \cdot \overline{C}$$

Trick.

$F = AB + C$

$\rightarrow F_d = (A + B) \cdot C$

$\rightarrow F_c = (\overline{A} + \overline{B}) \cdot \overline{C}$

Qus(2): Find the Compliment of the Expression

$$F = A(B+C)$$

Soln:

$$F = \overline{A \cdot (B+C)}$$

$$= \overline{A} + \overline{(B+C)}$$

$$= \overline{A} + \overline{B} \cdot \overline{C}$$

$$F_c = A + B \cdot C$$

$$F_d = A + B \cdot C$$

$$F_c = \overline{A} + \overline{B} \cdot \overline{C}$$

Que③: Find the complement of the Expression.

$$F = a[b + z(x + \bar{a})]$$

BODMAS.

Soln:

$$F_d = a + b \cdot [z + (x \cdot \bar{a})]$$

$$= \bar{a} + \bar{b} \cdot [\bar{z} + (\bar{x} \cdot \bar{\bar{a}})]$$

$\bar{\bar{a}} = a$

$$= \bar{a} + \bar{b} \cdot [\bar{z} + (\bar{x} \cdot a)]$$

$$= \bar{a} + \bar{b}\bar{z} + \bar{b}\bar{x} \cdot a$$

$$= \bar{a} + \bar{b}\bar{x} \cdot a + \bar{b}\bar{z}$$

$$= (\bar{a} + a)(\bar{a} + \bar{b}\bar{x}) + \bar{b}\bar{z}$$

$$\bar{F} = \bar{a} + \bar{b}\bar{x} + \bar{b}\bar{z}$$

Qus④: Find the compliment of the Expression

$$F = a(b+c) + \bar{a}b$$

soln:

$$F_d = [a+(b\cdot c)]\cdot[\bar{a}+b]$$

$A \cdot A = A$

$$\bar{F} = [\bar{a}+(\bar{b}\cdot\bar{c})][\bar{a}+\bar{b}]$$

$$= [\bar{a}+(\bar{b}\cdot\bar{c})][a+\bar{b}] \quad \text{---} \quad (2)$$

$$= \bar{a}\cdot a + \bar{a}\bar{b} + a\bar{b}\cdot\bar{c} + \bar{b}\cdot\bar{b}\cdot\bar{c}$$

$$= \bar{a}\bar{b} + a\bar{b}\cdot\bar{c} + \bar{b}\cdot\bar{c}$$

$$= \bar{b}[\bar{a}+a\bar{c}] + \bar{b}\bar{c}$$

$$= \bar{b}[\bar{a} + a\bar{c}] + \bar{b}\bar{c}$$

$$= \bar{b}[(\bar{a} + a)(\bar{a} + \bar{c})] + \bar{b}\bar{c}$$

$A + A = A$

$$= \bar{b}[(\bar{a} + \bar{c})] + \bar{b}\bar{c}$$

$$= b\bar{a} + bc + bc$$

$$= \bar{b}\bar{a} + \bar{b}\bar{c}$$

$$= \bar{b}(\bar{a} + \bar{c})$$

$$F = a(b+c) + \bar{a}b$$

$$F_d = [a + bc] \cdot [\bar{a} + b]$$

$$F_c = (\bar{a} + b\bar{c}) \cdot (\bar{a} + \bar{b})$$

$$= (\bar{a} + \bar{b}\bar{c})(a + \bar{b}) \quad \text{—— T2}$$

$$= \bar{a}\bar{b} + a\bar{b}\bar{c}$$

$$= \bar{b}[\bar{a} + a\bar{c}]$$

$$= \bar{b}[(\bar{a} + a)(\bar{a} + \bar{c})]$$

$$= \bar{b}[(\bar{a} + \bar{c})]$$

(Consensus theorem / Redundancy theorem)

→ 3 Term

→ find 1-variable in normal form and complement form.

$f = AB + \bar{A}C + BC$ — Eliminate.

→ A → (A) → normal form

→ $\bar{A}$ → complement form.

$$f = AB + \bar{A}C$$

proof:

$A + \bar{A} = 1$

$F = AB + \bar{A}C + BC$

$1 \cdot BC \rightarrow BC$

$1 \cdot AC \rightarrow AC$

$f = AB + \bar{A}C + 1 \cdot BC$

$= AB + \bar{A}C + (A+\bar{A})BC$

$= AB + \bar{A}C + ABC + \bar{A}BC$

$= ABC(1+C) + \bar{A}C(1+B)$

$= AB + \bar{A}C$

Qus②: $A\bar{B} + AC + BC$

Soln: $A\bar{B} + BC$

Qus③: $AB + AC + B\bar{C}$

Soln: $AC + B\bar{C}$

Qus: $F = \bar{A}\bar{B} + \bar{B}\bar{C} + \bar{A}C$

$F =$

$\bar{B}\bar{C} + \bar{A}C$

Qus④: $(A+B)\cdot(\bar{A}+C)(B+C)$

$(A+B)\cdot(\bar{A}+C)$

Qus⑤: $(A+B)(A+C)(B+\bar{C})$

$(A+C)\cdot(B+\bar{C})$

$\bar{A}, \bar{B}, C$ ✓

Que: $X * Y = \overline{XY}$, then minimized Expression of $[((x+y)*y)*z]$ is

$X * Y = \overline{XY}$

Soln:

$$(x+y)*y = \overline{(x+y).y}$$

$$= \overline{xy+y}$$

$$= \overline{y(x+1)}$$

$$= \overline{y}$$

$$\overline{y} * z = \overline{\overline{y}.z}$$

$$= \overline{\overline{y}} + \overline{z} \Rightarrow \underline{y+\overline{z}}$$

Que: If $f(A,B) = \bar{A}+B$ then the simplified expression of

$$f[f(x+y, y), z] \text{ is } \text{------}$$

$(\bar{x}+y)$

$f(A,B) = \bar{A}+B$

Soln:

$$F(x+y, y) = \overline{\bar{x}+y}+y$$

$A \quad B$

$$= \bar{x}\cdot\bar{y}+y$$

$$= (y+\bar{y})(y+\bar{x}) \Rightarrow (\bar{x}+y)$$

$$F[\underbrace{\bar{x}+y}_{A}, \underbrace{z}_{B}] = \overline{\bar{x}+y}+z$$

$$= \bar{\bar{x}}\cdot\bar{y}+z$$

$$= x\cdot y+z$$

Que: Minimize the following Boolean Expression.

$$(A+B+C)\,(A+B+\bar{C})\,(A+\bar{B}+C)$$

(1) (2) (3) (4)

T_1 → (1)(3) + (2)(4)

Soln:

$$((A+B)+C\cdot\bar{C})\,(A+\bar{B}+C)$$

$$(A+B)\,(A+\bar{B}+C)$$

$$(A+B(\bar{B}+C))$$

$$(A+B\cdot\bar{B}+BC)$$

$$= A+BC$$

Qus:

$F = xy + \overline{xy}\,wz$

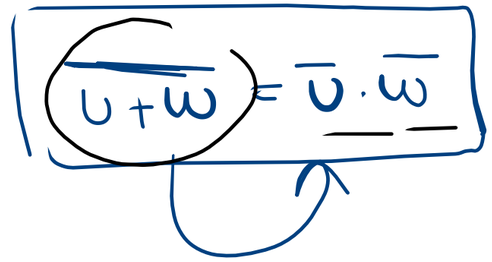
$F =$

$(xy + \overline{xy})(xy + wz)$

$F = (xy + wz)$

Qus: Minimize the following Boolean Expression.

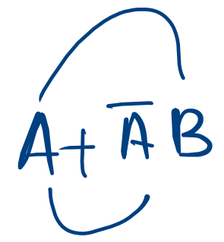
$$V + \bar{V}W + \bar{V}\bar{W}X + \bar{V}\bar{W}\bar{X}q$$



Soln:

$$(U+\bar{U})(U+W) + \bar{V}\bar{W}(X+\bar{X}q)$$

$$= (U+W) + \bar{V}\bar{W}((X+\bar{X})(X+q))$$



$$= (U+W) + \bar{U}\bar{W}(X+q)$$

$$= \underbrace{(U+W)}_{A} + \underbrace{\overline{(V+W)}}_{\bar{A}}\underbrace{(X+q)}_{B} \rightarrow \text{dual: } A+\bar{A}$$

$$= (U+W+\overline{U+W})(U+W+X+q)$$

$$= (U+W+X+q)$$

Qus: Minimize the following Boolean expression.

$$A + \bar{A}B + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C}D$$

Soln.

$$(A+B) + \bar{A}\bar{B}(C + \bar{C}D)$$

$$(A+B) + \bar{A}\bar{B}(C+D)$$

$$\underbrace{(A+B)}_{A} + \underbrace{\overline{(A+B)}}_{\bar{A}}\underbrace{(C+D)}_{B}$$

$$(A+B + \overline{A+B})(A+B+C+D)$$

A and $\bar{A}$

$$= (A+B+C+D)$$

$$\overline{A+B} = \bar{A}\cdot\bar{B}$$

$A + \bar{A}B$

$A + \bar{A} = 1$

Qus: minimize the Boolean Expression.

$$F = \bar{A}B + \bar{B}C + \bar{C}A + \bar{B}A + AC + B + \bar{C}$$

Sol^n:

$$F = B(\bar{A}+1) + \bar{C}(A+1) + \bar{B}C + \bar{B}A + AC$$

$$F = B + \bar{C} + \bar{B}C + \bar{B}A + AC$$

$$F = B + \bar{B}C + \bar{B}A + \bar{C} + AC$$

$$= B + \bar{B}C + \bar{B}A + A + \bar{C}$$

$$= B + C + \bar{B}A + A + \bar{C}$$

$$= 1 + \text{————} \Rightarrow 1$$

1 + Anything = 1

C + C̄ = 1

F = 1

Boolean function Representation.

① Canonical form → CSOP, CPOS

② Minimal form / standard form

$F(A,B,C) = (A\bar{B}C) + ABC + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C$ — CSOP

(Canonical Sum of product)

3-variable.

CSOP → Sum → 4 mint.

{ minterm → product term
use → sum.

$F(A,B,C) = (A+B+\bar{C}) \cdot (\bar{A}+\bar{B}+\bar{C}) \cdot (A+B+C) \cdot (\bar{A}+\bar{B}+C)$

Canonical form → CPOS

4-maxterm

{ maxterm → sumterm
use → product

① Canonical Form: Each minterm (maxterm) contains all the Boolean variables.

CSOP CPOS

minterm → '1'

$F(A,B,C) = \bar{A}BC + A\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C}$ → CSOP

$f(A,B,C) = (A+B+C)\cdot(\bar{A}+\bar{B}+\bar{C})(A+B+\bar{C})(A+\bar{B}+\bar{C})$

↳ CPOS

→ maxterm → '0'

② Minimal form (Standard form):

$f(A,B,C) = AB + AC$ → SOP form.

$f(A,B,C) = (A+\bar{B})\cdot(B+\bar{C})$ → POS form.

↳ the minimized form of Boolean Expression.

Literal: A Boolean variable either in Normal form (or) Complemented form is known as literal.

Minterm: Each term in Canonical SOP Representation is known as minterm. → '1'

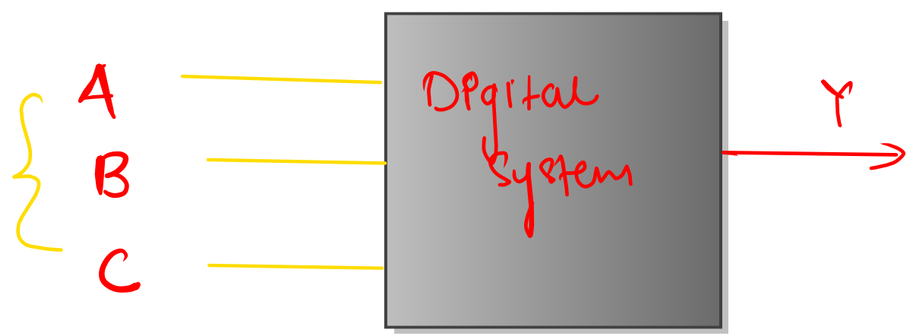
Maxterm: Each Term in Canonical POS Representation is known as maxterm. → '0'

$$f(A,B,C) = \bar{A}\bar{B} + \bar{B}\bar{C} + \bar{A}\bar{C}$$

6 - literal

n var — 2^n

→ (0 to $2^n - 1$)



↳ using 1 var no of possible input combinations $= 2^1 = 2$

using 2-var no of possible input combinations $= 2^2 = 4$

using 3-var " " " $= 2^3 = 8$

using n-var no of possible input combinations
$= 2^n$
(0 to $2^n - 1$)

SOP → logic '0' → $\bar{A}$

	A	B	C	Y
0 —	0	0	0	1
1 —	0	0	1	0
2 —	0	1	0	1
3 —	0	1	1	1
4 —	1	0	0	0
5 —	1	0	1	1
6 —	1	1	0	1
7 —	1	1	1	0

$\bar{A}\bar{B}\bar{C}$

$\bar{A}B\bar{C}$

3 var — $2^3 = 8$ input

max 8 minterms, maxterm = 0

$2^3 = 8$

(0 to 7)

minterm — '1'

maxterm — '0'

8-input combination

5 minterm

3 maxterm

4 var — $2^4 = 16$ (0 to 15)

0 to $2^n - 1$

8 minterm

8 maxterm

Sum of product (SOP) :-

0
1 ✓
2
3
4 ✓
5
6
7 ✓

Focus — minterm — '1'

(Focus on '1')

Logic '1' — A (Normal form)

Logic '0' — $\bar{A}$ (Complemented form)

C SOP form

$$Y(A,B,C) = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}C + ABC$$

0 0 0 — 0 1 0 — 0 1 1 — 1 0 1 — 1 1 1

(0) (2) (3) (5) (7)

$$Y(A,B,C) = \sum m(0,2,3,5,7)$$

$$m_0 + m_2 + m_3 + m_5 + m_7$$

② Product of Sum (POS) :

focus → maxterm → '0'

Logic '1' — $\bar{A}$ (complement form)

Logic '0' — A (normal form)

1 4 6

001 100 110

$F(A,B,C) = (A+B+\bar{C}) \cdot (\bar{A}+B+C) \cdot (\bar{A}+\bar{B}+C)$

001 !

$$f(A,B,C) = \prod M(1,4,6)$$

$$f(A,B,C) = M_1 \cdot M_4 \cdot M_6$$

$n = 3 \quad -2^3 = (8) \rightarrow (0 \text{ to } 2^3 - 1) \rightarrow (0 \text{ to } 7)$

A	B	C	Y
0	0	0	1 — 0
0	0	1	0 — 1
0	1	0	0 — 2
0	1	1	1 — 3
1	0	0	0 —
1	0	1	1 — 5
1	1	0	0 —
1	1	1	0

→ (8)

(3) (5)

minterm maxterm

$y(A,B,C) = \Sigma m(0,3,5)$

$= m_0 + m_3 + m_5$

$y(A,B,C) = \Pi M(1,2,4,6,7)$

n-var

(1) Number of minterm + no of maxtem = 2^n

(2) maximum possible minterms = 2^n, maxtem = 0

(3) maximum possible maxtems = 2^n, mintem = 0

(4) the Sum of All maximum possible minterms = 1

$$Y(A,B,C) = \Sigma m(0,1,2,3,4,5,6,7)$$

$$Y = \underset{000}{\bar{A}\bar{B}\bar{C}} + \underset{001}{\bar{A}\bar{B}C} + \underset{010}{\bar{A}B\bar{C}} + \underset{011}{\bar{A}BC} + \underset{100}{A\bar{B}\bar{C}} + \underset{101}{A\bar{B}C} + \underset{110}{AB\bar{C}} + \underset{111}{ABC}$$

$$y = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC$$

$$= \bar{A}\bar{B}(\bar{C}+C) + \bar{A}B(\bar{C}+C) + A\bar{B}(\bar{C}+C) + AB(\bar{C}+C)$$

$$= \bar{A}B + AB + AB + A\bar{B}$$

$$= \bar{A}(\bar{B}+B) + A(\bar{B}+B)$$

$$= \bar{A} + A$$

$$y = 1$$

$$y = \sum_{i=0}^{2^n-1} m_i = 1$$

n-var

(0 to 2^n-1)

(5) The product of All possible maxterms = 0

$2^2 = 4$ (0-3)
(0,1,2,3)

$Y(A,B) = \prod M(0,1,2,3)$ — logic '0' — A
logic '1' — $\bar{A}$

$Y(A,B) = (A+B)\cdot(A+\bar{B})\cdot(\bar{A}+B)\cdot(\bar{A}+\bar{B})$

$= (A+B\bar{B})(\bar{A}+B\cdot\bar{B})$

$= A\cdot\bar{A} = 0$

$= \prod_{i=0}^{2^n-1} M_i = 0$

qmp (6) Minterms and maxterms are of same index are complement to Each other.

$m_5 = \underset{A\bar{B}C}{101}$ $\quad \overline{A\bar{B}C} \longrightarrow (\bar{A}+\bar{\bar{B}}+\bar{C})$

$\bar{m}_5 = M_5 \longrightarrow (\bar{A}+B+\bar{C})$ $\quad (\bar{A}+B+\bar{C})$

$M_5 = m_5$

$(\bar{A}+B+\bar{C}) \Rightarrow \bar{\bar{A}}.\bar{B}.\bar{\bar{C}} \Rightarrow \underset{A.\bar{B}.C}{101} \rightarrow m_5$

$$\begin{cases} \overline{m_{10}} = M_{10} \\ \overline{m_{87}} = M_{87} \end{cases}$$

QnP (7)

the product of two minterms of different index is = 0

f(A,B)

$m_2 \cdot m_3 = 0 \longrightarrow (A\bar{B}) \cdot (A B)$

$= A \cdot A \cdot \bar{B} B$

$= 0$

$m_2 \cdot m_2 = m_2$ $\qquad A \cdot A = A$

$(A\bar{B}) \cdot (A\bar{B})$

$= A\bar{B} \longrightarrow m_2$

$$= \begin{cases} m_i \cdot m_j = 0, & i \neq j \\ m_i, & i = j \end{cases}$$

(8) the sum of two arbitrary maxterms of diffent indexi's ----

$$M_2 + M_3 = (\bar{A}+B) + (\bar{A}+\bar{B})$$

$$= \bar{A}+B+\bar{A}+\bar{B}$$

$$= \bar{A}+1$$

$$= 1$$

$$M_2 + M_2 = (\bar{A}+B) + (\bar{A}+B)$$

$$= (\bar{A}+B) \rightarrow M_2$$

$$\{ M_i + M_j = 1 \text{ , } i \neq j \}$$

$$\{ M_i \text{ , } i = j \}$$

Qus: Find the minterms and maxterms of the following

$F(A,B,C) = AB + \bar{A}BC + C$ — SOP — CSOP

Method ①

$2^3 \rightarrow 8$ (0 to 7)

0
1
2
3
4
5
6
7

$F(A,B,C) = AB.1 + \bar{A}BC + 1.1.C$

$= AB(C+\bar{C}) + \bar{A}BC + (A+\bar{A})(B+\bar{B}).C$

$= ABC + AB\bar{C} + \bar{A}BC + (AB + A\bar{B} + \bar{A}B + \bar{A}\bar{B}).C$

$= ABC + AB\bar{C} + \bar{A}BC + ABC + A\bar{B}C + \bar{A}BC + \bar{A}\bar{B}C$

normal form
A → logic '1'
$\bar{A}$ → logic '0'

$= ABC + AB\bar{C} + \bar{A}BC + A\bar{B}C + \bar{A}\bar{B}C$ — CSOP

1 1 1 | 1 1 0 | 0 1 1 | 1 0 1 | 0 0 1

$f(A,B,C) = \Sigma m(1, 3, 5, 6, 7)$ — minterms

minterm → maxterm

POS — CPOS

$f(A,B,C) = \Pi M(0,2,4)$

Method ②: MSB → LSB

$f(A,B,C) = AB + \bar{A}BC + C$ — SOP

Shortcut

AB —	$\bar{A}BC$	— — C
6 — 1 1 0 → $\bar{C}$	0 1 1 — 3	0 0 1 — 1
7 — 1 1 1 → C		0 1 1 — 3
		1 0 1 — 5
		1 1 1 — 7

SOP → CSOP → minterm

$$\begin{cases} f(A,B,C) = \Sigma m(1,3,5,6,7) \\ f(A,B,C) = \Pi M(0,2,4) \end{cases}$$

Qus ② Find the minterms and maxterms of the following

$$F(A,B,C) = (B+C)\cdot(\bar{A})\cdot(\bar{A}+\bar{B}+C) \longrightarrow \text{POS} \rightarrow \text{CPOS} \rightarrow \text{maxterms} \rightarrow \text{minterm}$$

logic '1' → $\bar{A}$

logic '0' → A

Sol:

	$-+B+C$	$\bar{A}+-+-$	$\bar{A}+\bar{B}+C$
0 —	0 0 0		
4 —	1 0 0	1 0 0 — 4	
5 —		1 0 1 — 5	
6 —		1 1 0 — 6	1 1 0 — 6
7 —		1 1 1 — 7	

$f(A,B,C) = \prod M(0,4,5,6,7)$ — maxterms

$f(A,B,C) = \Sigma m(1,2,3)$

Qus: Find the minterms and maxterms of the following

$f(A,B,C) = \bar{A}B + A\bar{C} + B + ABC$ — SOP

Sol^n:

	$\bar{A}B$	$A\bar{C}$	B	ABC
2 —	0 1 0	1 0 0 -4	0 1 0 -2	1 1 1 -7
3 —	0 1 1	1 1 0 -6	0 1 1 -3	
			1 1 0 -6	
			1 1 1 -7	

$f(A,B,C) = \Sigma m(2,3,4,6,7)$ —

$f(A,B,C) = \Pi M(0,1,5)$

Qus ③

$f(A,B,C) = A + B + \bar{A}B + C$ — SOP

	A	B	$\bar{A}B$	C
4 —	1 0 0			
2 —		0 1 0	0 1 0 — 2	
5 —	1 0 1			
3 —		0 1 1	0 1 1 — 3	
6 —	1 1 0			
6 —		1 1 0		
7 —	1 1 1			
7 —		1 1 1		
				0 0 1 — 1
				0 1 1 — 3
				1 0 1 — 5
				1 1 1 — 7

$f(A,B,C) = \Sigma m(1,2,3,4,5,6,7)$ —

↳ $m_1 + m_2 + m_3 + m_4 + m_5 + m_6 + m_7$

$f(A,B,C) = \Pi M(0)$

↳ M_0

Qus④: $F(A,B,C) = A+B+C$ — SOP — CSOP — minterm / maxterm

	A	B	C	
4	1	0	0	
5	1	0	1	
6	1	1	0	
7	1	1	1	
		0	1	0	-2
		0	1	1	-3
		1	1	0	-6
		1	1	1	-7
			0	0	1	-1
			0	1	1	-3
			1	0	1	-6
			1	1	1	-7

$f(A,B,C) = \Sigma m(1,2,3,4,5,6,7)$

$f(A,B,C) = \Pi M(0)$

$f(A,B,C) = (A+B+C)$ — POS → (POS

0 0 0 — 0

↓ maxterm

↓ minterm

$f(A,B,C) = \Pi M(0)$

$f(A,B,C) = \Sigma m(1,2,3,4,5,6,7)$

minterms & maxterms = ?

Ques: $f(A,B,C,D) = AB + \bar{A}BC + B\bar{C}$ — SOP

Soln: $2^4 = 16$

(0 to 15)

$0 \text{ to } 2^n - 1$

$0 \text{ to } 2^4 - 1$

(0 to 15)

AB — —
1 1 0 0 — 12
1 1 0 1 — 13
1 1 1 0 — 14
1 1 1 1 — 15

$\bar{A}BC$ —
0 1 1 0 — 6
0 1 1 1 — 7

— $B\bar{C}$ —
0 1 0 0 — 4
0 1 0 1 — 5
1 1 0 0 — 12
1 1 0 1 — 13

$f(A,B,C,D) = \Sigma m(4,5,6,7,12,13,14,15)$

$f(A,B,C,D) = \Pi M(0,1,2,3,8,9,10,11)$

Qus: $f(A,B,C) = (\bar{B}+\bar{C})\cdot(\bar{A}+\bar{B})\cdot(A+B+C)(A)$ — pos → Cpos → maxterm → minterm.

$\bar{B}+\bar{C}$ — + —

0	1	1	-3
1	1	1	-7

$\bar{A}+\bar{B}+-$

1	1	0	-6
1	1	1	-7

$A+B+C$

0	0	0	-0

$A+-+-$

0	0	0	-0
0	0	1	-1
0	1	0	-2
0	1	1	-3

$f(A,B,C) = \Pi M(0,1,2,3,6,7)$

$f(A,B,C) = \Sigma m(4,5)$

Qus: $f(P,Q,R) = \overline{P} + QR$. — SOP — CSOP — minterm — maxterm

$\overline{P}$				QR	
0	0	0	0	0 1 1	3
0	0	1	1	1 1 1	7
0	1	0	2		
0	1	1	3		

$\Sigma m(0,1,2,3,7)$

$\pi M(4,5,6)$

Qus: $F(P,Q,R,S) = PQ + \bar{P}QR + \bar{P}Q\bar{R}S$ — SOP — CSOP - min -

PQ				$\bar{P}QR$				$\bar{P}Q\bar{R}S$			
1	1	0	0	0	1	1	0	0	1	0	1
1	1	0	1	0	1	1	1				
1	1	1	0								
1	1	1	1								

-12 -13 -14 -15 | -6 -7 | -5

$f(P,Q,R,S) = \sum m(5,6,7,12,13,14,15)$

$f(P,Q,R,S) = \prod M(0,1,2,3,4,8,9,10,11)$

Qus: $F_1(A,B,C) = \Sigma m(2,3,6)$ and $F_2(A,B,C) = \Sigma m(1,2,5,7)$

then $F_3 = F_1 \cdot F_2$ $\quad$ $F_4 = F_1 + F_2$

Soln:

$$F_3 = F_1 \cdot F_2$$

$$F_1(2,3,C) \cdot F_2(1,2,5,7)$$

$$(m_2 + m_3 + m_6) \cdot (m_1 + m_2 + m_5 + m_7)$$

$$F_3 = m_2 \Rightarrow \Sigma m(2)$$

$$F_4 = F_1 + F_2$$

$$= (m_2 + m_3 + m_6) + (m_1 + m_2 + m_5 + m_7)$$

$$F_4 = m_2 + m_3 + m_6 + m_5 + m_7 + m_1$$

$F_3 = \Sigma m(2)$

$F_4 = \Sigma m(1, 2, 3, 5, 6, 7)$

$m_i \cdot m_j = 0 \quad , \quad i \neq j$

$m_i \quad , \quad i = j$

$m_i \cdot m_i = m_i$

Qus: $F(P,Q,R,S) = PQ + \bar{P}QR + \bar{P}Q\bar{R}S$, the function is equivalent to

a) $PQ + QR + QS$

b) $P + Q + R + S$

c) $\bar{P} + \bar{Q} + \bar{R} + \bar{S}$

d) $\bar{P}R + \bar{P}\bar{R}S + P$

Soln:

$$F = PQ + \bar{P}QR + \bar{P}Q\bar{R}S$$

$$= Q(P + \bar{P}R + \bar{P}\bar{R}S)$$

$$= Q(P + R + \bar{P}\bar{R}S)$$

$$= Q(R + P + \bar{P}\bar{R}S)$$

$$= Q(R + P + \bar{R}S) \Rightarrow$$

$$Q(P + R + \bar{R}S)$$

$$Q(P + R + S)$$

$P + \bar{P}\bar{R}S$

$(P+\bar{P})(P+\bar{R}S)$

$R + \bar{R}S$

$(R+\bar{R})(R+S)$

$$F = \underline{QP + QR + QS}$$

$F(A,B,C) = \Sigma m(1,2,3,4) \longrightarrow 0,5,6,7$

$\overline{F}(A,B,C) = \Sigma m(0,5,6,7)$

$\overline{F}(A,B,C) = \Pi M(0,5,6,7)$

Qus: Find the minterms of $\overline{F}$

$F(A,B,C) = AB + \overline{A}BC + C$ — SOP — CSOP — minterms (F) ↓ $\overline{F}$ = ?

	AB—	$\overline{A}$BC	——C	
6—	1 1 0	0 1 1	0 0 1	—1
7—	1 1 1	1	0 1 1	—3
		2	1 0 1	—5
			1 1 1	—6

$f(A,B,C) = \Sigma m(1,3,5,6,7)$

$f(A,B,C) = \Pi M(0,2,4)$

$\overline{f}(A,B,C) = \Sigma m(0,2,4)$

Ques: Using two Boolean variables, How many different Boolean functions are possible.

2-var $= 2^2 = 4$

Soln:

$f(A,B) = \Sigma m(0)$

$f(A,B) = \Sigma m(1)$

$f(A,B) = \Sigma m(2)$

$f(A,B) = \Sigma m(3)$

$f(A,B) = \Sigma m(0,1)$

$f(A,B) = \Sigma m(0,2)$

$f(A,B) = \Sigma m(0,3)$

$f(A,B) = \Sigma m(0,1,2)$

$f(A,B) = \Sigma m(0,1,3)$

2^{2^n} → input combination.

$2^{(2^2)} \Rightarrow 2^4 \Rightarrow 16$ (0 to 15)

A	B	f_0	f_1	f_2	f_3	f_4	f_5	f_6	f_7	f_8	f_9	f_{10}	f_{11}	f_{12}	f_{13}	f_{14}	f_{15}
0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
0	1	0	0	0	0	1	1	1	1	0	0	0	0	1	1	1	1
1	0	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1
1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1

$n = 2, \quad 2^2 = 4$ input combination.

$2^{2^n} = \quad 2^{2^2} = 16$

3 var $= \quad 2^3 = 8$

$2^{2^3} = \quad 2^8 = \underline{256}$

using n-var How many functions are possible.

n-var → 2^n = 2^{2^n}

How many Boolean Expression possible with 1 minterm?

n = 2 , $2^2 = 4$ input combination

with 0 minterm $4C_0$ | 1 minterm $4C_1$ | 2 minterm $4C_2$ | 3 minterm $4C_3$ | 4 minterm $4C_4$

$n = 3$, $2^3 = 8$, $2^{2^3} = 256$ function.

0 minterm: $^8C_0$

1 minterm: $^8C_1$

2 minterm: $^8C_2$

$^8C_3$ $^8C_4$ $^8C_5$ $^8C_6$ $^8C_7$

$^8C_8$

$n=2$

Qus: By using 2-Boolean variables total no of possible Boolean function = $2^{2^n} = 2^{2^2} = 16$

Qus(2): By using n-Boolean variables total number of possible Boolean function = 2^{2^n}.

Qus: By Using 2-Boolean variables total number of possible functions having Atmost 3-minterms

(0+1+2+3)

Sol^n:

- Atleast → (3) → min ≥ (3), 4, 5, 6 - - - - - - - - - -
- Atmost → maxm → (3) → $x \geq 3$

0, 1, 2, 3

$x \leq 3$

0+1+2+3

$P(x \leq 3)$ $\quad n=2, \quad 2^2 = 4$

0+1+2+3

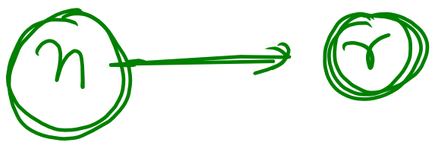
$\Rightarrow \quad {}^4C_0 + {}^4C_1 + {}^4C_2 + {}^4C_3$

$\Rightarrow$ (15)

$${}^nC_r = \frac{n!}{(n-r)! \times r!}$$

Qus: By using 2- Boolean variables total number of possible Boolean functions having At most 3-maxterms:

Soln: $P(x \leq 3)$



minterm.

$n \hookrightarrow (n-r)$

$\rightarrow {}^{n}C_{r} = {}^{n}C_{(n-r)}$

$= \boxed{{}^{n}C_{r} = {}^{n}C_{(n-r)}}$

$= {}^{4}C_{0} + {}^{4}C_{1} + {}^{4}C_{2} + {}^{4}C_{3}$

$= 15$

Qus(5): By using 2- Boolean variables total number of Possible Boolean function having 3-minters $\Rightarrow {}^4C_3 \Rightarrow (4)$

Qus(6): By using n- Boolean variables total Number of possible Boolean function having 2-minterms

n-var $\rightarrow (2^n)$

$= {}^{2^n}C_2$

$n=5,\ 2^5 = (32)$

Qus(7): By using 5- Boolean variables total number of possible Boolean functions having Atmost 3 minterms $({}^{n}C_3) \rightarrow (0+1+2+3)$

$= {}^{32}C_0 + {}^{32}C_1 + {}^{32}C_2 + {}^{32}C_3$

$= 5489.$

Qus: $Y(A,B,C) = \Sigma m(1,4,6,7)$ Qualify the Correct stmts.

$Y(A,B,C) = \Sigma m(1,4,6,7)$ $\quad$ $Y(A,B,C) = \Pi M(0,2,3,5)$

$= m_1 + m_4 + m_6 + m_7$ $\quad$ $= M_0 \cdot M_2 \cdot M_3 \cdot M_5$

True

(1) $Y = m_1 + m_4 + m_5 + m_7 \rightarrow (1,4,6,7)$

(2) $Y = M_0 M_2 M_3 M_5 \rightarrow$ True

(3) $Y = \overline{m_0 + m_2 + m_3 + m_5} \rightarrow$ True.

$= \overline{m_0} \cdot \overline{m_2} \cdot \overline{m_3} \cdot \overline{m_5}$

$Y = M_0 \cdot M_2 \cdot M_3 \cdot M_5 \rightarrow (0,2,3,5)$

(4) $Y = \overline{M_1 \cdot M_4 \cdot M_6 \cdot M_7} \Rightarrow$

$\overline{M_1} + \overline{M_4} + \overline{M_6} + \overline{M_7} \Rightarrow m_1 + m_4 + m_6 + m_7 \rightarrow$ True

1

(5) $Y = \overline{m_0} \cdot \overline{m_2} \cdot \overline{m_3} \cdot \overline{m_5}$ → True

$Y = M_0 \cdot M_2 \cdot M_3 \cdot M_5 \rightarrow (0, 2, 3, 5)$

(6) $Y = \overline{M_1} + \overline{M_4} + \overline{M_6} + \overline{M_7}$ → True.

$Y = m_1 + m_4 + m_6 + m_7$

false

(7) $Y = m_0 \cdot m_2 \cdot m_3 \cdot m_5$

$\boxed{Y = 0}$

false

(8) $Y = M_1 + M_4 + M_6 + M_7 \Rightarrow (1)$

$Y = 1$

(9) $Y = \overline{m_1 \cdot m_4 \cdot m_6 \cdot m_7}$

$= \overline{m_1} + \overline{m_4} + \overline{m_6} + \overline{m_7}$

$Y = M_1 + M_4 + M_6 + M_7$

false

false

(10) $Y = \overline{M_0 + M_2 + M_3 + M_5}$

$= \overline{M_0} \cdot \overline{M_2} \cdot \overline{M_3} \cdot \overline{M_5}$

$Y = m_0 \cdot m_2 \cdot m_3 \cdot m_5$

Qus: If $A * B = AB + \bar{A}\bar{B}$, Let $C = A * B$, then which of the Following is correct.

a) $B * C = A$ b) $A * C = B$ c) $A * B * C = 1$

d) $A * B = B * A$

Sol: $A * B = AB + \bar{A}\bar{B}$, $C = A * B = AB + \bar{A}\bar{B}$

Try a) $B * C = A$

LHS: $B * C = BC + \bar{B}\bar{C}$

$= B[AB + \bar{A}\bar{B}] + \bar{B}\overline{[AB + \bar{A}\bar{B}]}$

$= AB + \bar{B}[(\bar{A}\bar{B}) \cdot \overline{(\bar{A} \cdot B)}]$

$= AB + \bar{B}[(\bar{A} + \bar{B}) \cdot (A + B)]$

$$= AB + \bar{B}\left[(\bar{A}+\bar{B})\cdot(A+B)\right]$$

$$\xrightarrow{T_2} \quad (1)(4) + (2)(3)$$

$$= AB + \bar{B}\left[\bar{A}B + \bar{B}A\right]$$

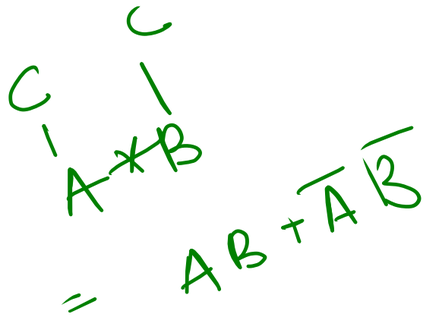
$$= AB + \bar{A}B\cdot\bar{B} + \bar{B}A\cdot\bar{B}$$

$$= AB + A\bar{B}$$

$$= A(B+\bar{B})$$

$$= (A)$$

$C = A * B$



$= AB + \bar{A}\bar{B}$

(c) $A * B * C = 1$

$C * C = 1$

LHS: $C * C = C \cdot C + \bar{C} \cdot \bar{C}$

$= C + \bar{C}$

$= 1$

Correct

(a) $A * B = B * A$

LHS: $A * B = AB + \bar{A}\bar{B}$

RHS: $B * A = B \cdot A + \bar{B}\bar{A}$

$= AB + \bar{A}\bar{B}$

AND- commutative law

↓

follow

$AB = BA$

(b) $A * C = B$

$C = AB + \bar{A}\bar{B}$

True

LHS: $A * C = AC + \bar{A}\bar{C}$

$= A[AB + \bar{A}\bar{B}] + \bar{A}[\overline{AB + \bar{A}\bar{B}}]$

$A \cdot \bar{A} = 0$

$= A \cdot AB + A \cdot \bar{A}\bar{B} + \bar{A}[(\bar{A}B + \bar{B}A)]$

$= AB + \bar{A}[AB + BA]$

$= AB + \bar{A} \cdot \bar{A}B + \bar{A} \cdot \bar{B} \cdot A$

$= AB + \bar{A}B$

$= B(A + \bar{A})$

$= (B)$

Neutral function.

If the number of minterms and number of maxterms are equal, then the Boolean function is called as Neutral function.

Neutral function

$$f(A,B,C) = AB + BC + AC$$ — SOP — CSOP — —

	A B —	— B C	A — C
6 —	1 1 0	0 1 1 — 3	1 0 1 — 5
7 —	1 1 1	1 1 1 — 7	1 1 1 — 7

$\Sigma m(3,5,6,7)$
Carry of Full Adder
majority of '1'

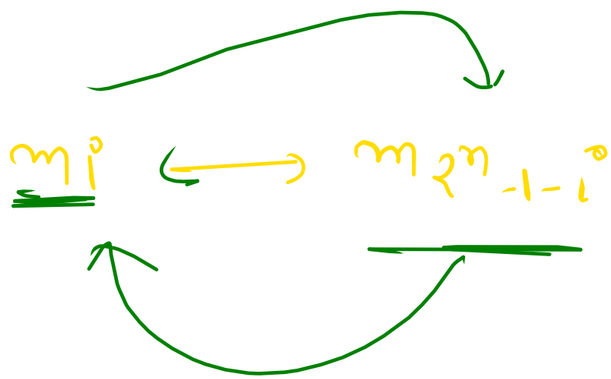
$$f(A,B,C) = \Sigma m(3,5,6,7) \quad \text{—— (4)}$$

$$f(A,B,C) = \Pi M(0,1,2,4) \quad \text{—— (4)}$$

(MET)

Mutually Exclusive terms:

the mutually exclusive term of m^i is $= m_{2^n-1-i^\circ}$



var $n=3$.

m_4

$i=4$

m_{2^3-1-4}

m_3

$m_4 \longleftrightarrow m_3$

$n=3$

m_3

$i^\circ=3$

m_{2^3-1-3}

m_4

$m_3 \longleftrightarrow m_4$

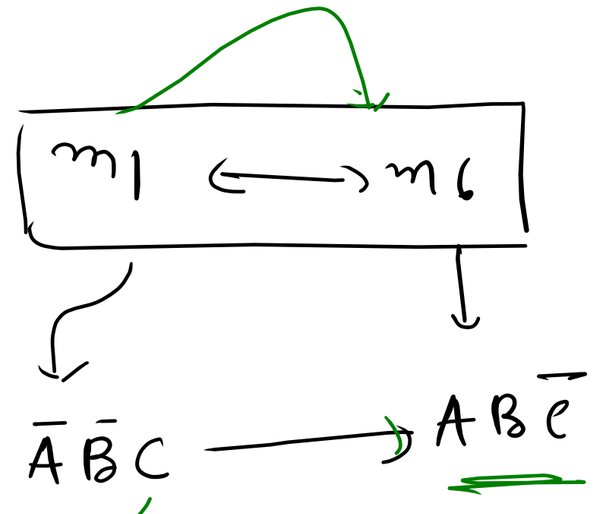
$n = 3$

m_1

$i = 1$

$m_{23} - 1 - 1$

m_6



m_3

$\downarrow$

$\bar{A}BC \longrightarrow A\bar{B}\bar{C}$

$1\ 0\ 0 \rightarrow m_4$

(4)

$\rightleftharpoons$

$m_3 \longleftrightarrow m_4$

$A\bar{B}\bar{C}\bar{D}E\bar{F}G$

↓ mutually Exclusive term.

$\bar{A}BCD\bar{E}F\bar{G}$

Var $n=2$ (0 to 3)

m_1, $i^0=1$

$\bar{A}B$ → 01 (m_1)

↓

1 0 → $A\bar{B}$ (m_2)

$m_1 \longleftrightarrow m_2$

$m_{2^n-1-i^0}$

m_{4-1-1}

m_2

$m_2 \longleftrightarrow m_1$

$f(A,B) \longrightarrow \boxed{n=2}$

Neutral function:

{ no of minterms = no of maxterms.

$2^n C_{2^{n-1}}$

$2^2 C_{2^{2-1}} \Rightarrow {}^4C_2$ } → 6

$= 6$

$\frac{4!}{2! \times 2!} \Rightarrow \frac{4 \times 3 \times 2!}{2 \times 1 \times 2!}$

→

A	B	f_1	f_2	f_3	f_4	f_5	f_6
0	0	0	1	0	1	1	0
0	1	0	1	1	0	0	1
1	0	1	0	0	1	0	1
1	1	1	0	1	0	1	0

Neutral function Application:

n-var → 2^n

no of minterms + no of maxterms $= 2^n$

x $\quad$ x

$$2x = 2^n$$

$$x = \frac{2^n}{2} = 2^{n-1}$$

$$\boxed{x = 2^{n-1}}$$

no of minterms $= 2^{n-1}$

no of maxterm $= 2^{n-1}$

Mutually Exclusive Term Application:

(1)

$$m_i \longrightarrow m_{2^n-1-i^0}$$

$n = 3$

ABC $\longleftrightarrow$ MBT ABC

$n = 3$, $2^3 = 8$ (0 to 7)

$i^0 = 0$, $m_0 \longleftrightarrow m_7$

$i^0 = 1$, $m_1 \longleftrightarrow m_6$

$i^0 = 2$, $m_2 \longleftrightarrow m_5$

$i = 3$, $m_3 \longleftrightarrow m_4$

0
1
2
3
4
5
6
7

(0,7)(1,6)(2,5)(3,4)

$n = 4, \quad 2^4 = 16 \quad (0 \text{ to } 15)$

Pair: { (0,15) (1,14) (2,13) (3,12) (4,11) (5,10) (6,9) (7,8)

Self Dual Expression.

$F = AB + BC + AC$ — Self Dual Expression

T_1

$f_d = (A+B)\cdot(B+C)\cdot(A+C)$ → ①③ + ②④

$f_d = (A+BC)(B+C)$

$f_d = AB + AC + BC\cdot B + BC\cdot C$

$f_d = AB + Ac + BC + BC$

$\boxed{f_d = AB + AC + BC}$

$\boxed{f = f_d}$

Both are equal.

If one time dual of the Boolean expression Result the same expression, then it is called as self dual expression.

Eg: $F = AB + BC + AC$ — $\Sigma m(3,5,6,7)$

$=$

neutral function

{ Self Dual Expression
  majority of logic
  Carry of full Adder.

Condition for the given expression is self Dual:

① the given Boolean function must be Neutral function.
i.e the number of minterms = no of maxterms $= 2^{n-1}$

no of minterms + no of maxterms $= 2^n$.

② It should not contain mutually exclusive terms
i.e if m_i belongs F, then m_{2^n-1-i} should belongs to $\overline{F}$.

$m_i \longleftrightarrow m_{2^n-1-i}$

$F(A,B,C) = \Sigma m(3,5,6,7)$

$F(A,B,C) = \Pi M(0,1,2,4)$

$\overline{F}(A,B,C) = \Sigma m(0,1,2,4)$

Qus: Verify the given functions are self dual or not

$$f(A,B,C) = AB + BC + AC$$ — Self Dual Expression

Sol: → $f(A,B,C) = \Sigma m(3,5,6,7)$ — (4)

$\overline{f}(A,B,C) = \Pi M(0,1,2,4)$ — (4)

→ $\overline{f}(A,B,C) = \Sigma m(0,1,2,4)$

① Neutral function: Equal

② $n = 3$, $2^3 = 8$ (0 to 7)

pair: (0,7) (1,6) (2,5) (3,4)

$\overline{f}$ f $\overline{f}$ f $\overline{f}$ f F $\overline{f}$

Ques②: Verify the given functions are self dual or not

Self dual expression

$\longrightarrow f(A,B,C) = \Sigma m(1,2,4,7)$ —— (4)

Sol^n: $f(A,B,C) = \Pi M(0,3,5,6)$ —— (4)

$\longrightarrow f(A,B,C) = \Sigma m(0,3,5,6)$

① Neutral function:

② $n = 3,\ 2^3 = 8$ (0 to 7)

Pair: (0,7) (1,6) (2,5) (3,4)

↓	↓	↓	↓	↓	↓	↓	↓
F	$\bar{F}$	F	$\bar{F}$	F	$\bar{F}$	$\bar{F}$	F

Que ③: Verify the given function are self Dual or not

$$f(A,B,C) = \Sigma m(0,1,2,5) \quad \text{—— (4)}$$

Soln:

$$f(A,B,C) = \Pi M(3,4,6,7) \quad \text{—— (4)}$$

$$\overline{f}(A,B,C) = \Sigma m(3,4,6,7)$$

① Equal.

② $n = 3, \quad 2^3 = 8, \quad (0 \text{ to } 7)$

pair: (0,7) (1,6) (2,5) (3,4)

f $\overline{f}$ f $\overline{f}$ f $\overline{f}$

not Self dual Expression.

Note !

1) Number of Boolean functions $= 2^{2^n}$

② Maximum no of minterms $= 2^n$

③ Maximum no of maxterms $= 2^n$

④ No of Self Dual Expression $= 2^{2^{n-1}}$

$n = 2$

$= 2^{2^{2-1}}$

$= 2^{2^1}$

$= 4$

⑤ Number of Neutral function $= {}^{2^n}C_{2^{n-1}}$

Qus: A logic Circuit have 3-inputs A, B, C and output Y is logic 1 for the following

Y → logic '1'

1) A and C are true
2) B and C are false
3) A, B and C are true
4) A, B and C are false.

then the minimized expression Y is ------

Soln:

$$Y = A \cdot C + \bar{B} \cdot \bar{C} + A \cdot B \cdot C + \bar{A} \cdot \bar{B} \cdot \bar{C}$$

$$= AC(1+B) + \bar{B}\bar{C}(1+\bar{A})$$

$$\longrightarrow \boxed{Y = AC + \bar{B}\bar{C}}$$

Qus ② A logic Circuit have 3-inputs A, B and C output is (F). F is logic '1' when majority number of Inputs are at logic '1', then the minimized Expression for F is.

Soln:

(F) → High → logic '1'

majority no of inputs

↓

logic '1' (High)

	A	B	C	F
→	0	0	0	0
→	0	0	1	0
→	0	1	0	0
→	0	1	1	1 → ③
→	1	0	0	0
→	1	0	1	1 → 5
→	1	1	0	1 → 6
	1	1	1	1 → 7

$f = \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$

$\bar{A}BC + ABC$

$f = BC(\bar{A}+A) + AC(B+\bar{B}) + AB(C+\bar{C})$

$f = BC + AC + AB$

$$f(A,B,C) = AB + BC + AC$$

→ majority logic('1')

→ minority logic('0')

→ $\Sigma m(3,5,6,7)$

→ self dual expression

→ carry of full Adder.

$\Sigma m(3,5,6,7)$

Qus: (3) A Car Alarm system is designed considering 4 inputs, Door closed (D), key in (K), Seat pressure (S), and Seat Belt closed (B), the Alarm (A) should sound if

(1) the key is in and door is not closed (or)

(2) the door is closed the key is in, driver in the seat, Belt is not closed then the minimized Expression is.

Soln:

$$A = K.\bar{D} + D.K.S\bar{B}$$

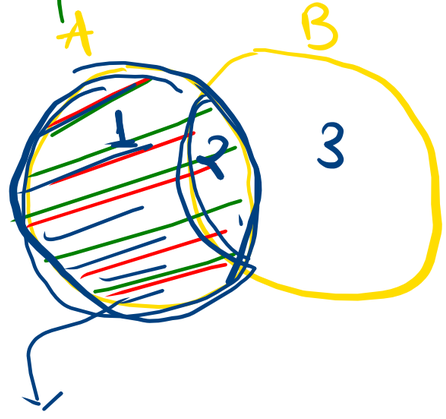
$$= K[\bar{D} + DS\bar{B}]$$

$$= K[(\bar{D}+D)(\bar{D} + S\bar{B})]$$

$$K[\bar{D} + S\bar{B}]$$

$$A = K\bar{D} + KS\bar{B}$$

Qus: for the given venn diagrams, find the minimized logical Expression.



Shaded means = 1

not shaded = 0

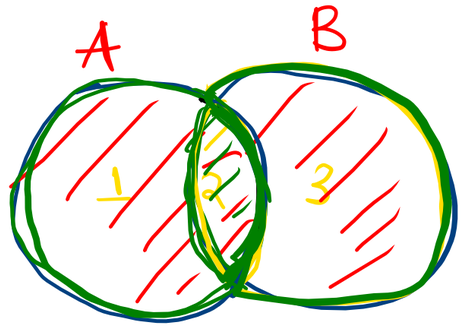
Sol^n:

$$= AB + A\bar{B}$$

$$A(B + \bar{B})$$

$$= A$$

Qus②:



→ A ∪ B

→ (A + B)

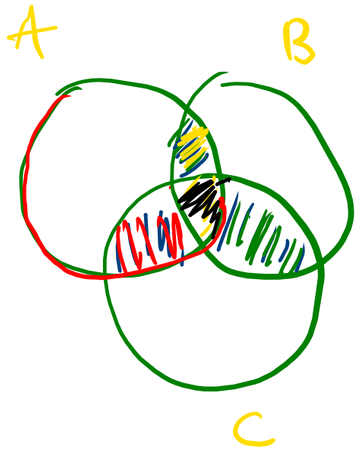
Soln: $F = A\cdot\overline{B} + AB + \overline{A}\cdot B$

$= A(B+\overline{B}) + \overline{A}\cdot B$

$= A + \overline{A}B$

$= (A+B)$

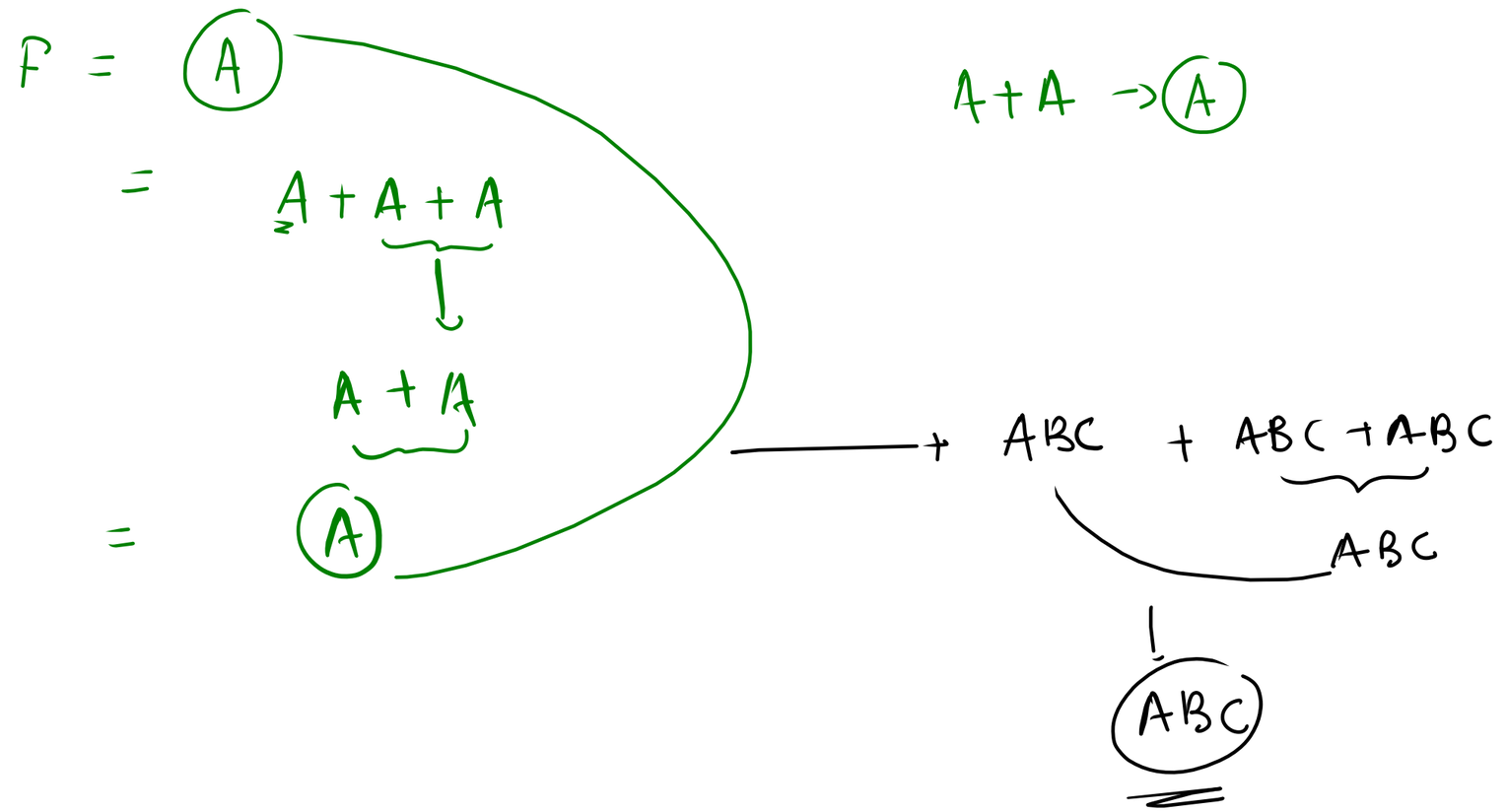
Qus (3)



$$F = A\cdot B\cdot C + \bar{A}\cdot B\cdot C + A\bar{B}\cdot\bar{C} + \cancel{A}BC + ABC + ABC$$

$$= AC(\bar{B}+B) + BC(\bar{A}+A) + AB(C+\bar{C})$$

$$\boxed{F = AC + BC + AB} \longrightarrow 3, 5, 6, 7$$


F = A
= A + A + A
A + A
= A
A + A → A
ABC + ABC + ABC
ABC
!
ABC